

Introduction to Neural Networks

Subject: Machine Learning

What is a Neural Network?

A neural network is a computing system loosely modeled after the human brain. It consists of layers of interconnected nodes (neurons) that process data through forward propagation and refine their weights through backpropagation during training.

Key Concepts

- Perceptron: A single-layer linear classifier that can separate linearly separable data.
- Activation Functions: Non-linear functions applied at each node (ReLU, Sigmoid, Tanh).
- Layers: Input layer, hidden layers, and output layer.
- Backpropagation: Gradient-based algorithm that propagates error backwards to update weights.
- Dropout: A regularization technique where random neurons are ignored during training to prevent overfitting.

Important Formulas

Linear transformation: $z = Wx + b$

Activation: $a = f(z)$

Cross-entropy loss: $L = -\sum y \cdot \log(\hat{y})$

Gradient descent: $W = W - \alpha \cdot (dL/dW)$

Exam Topics

1. Explain how backpropagation updates weights in a neural network.
2. What is the role of activation functions and why are non-linear ones needed?
3. Describe the difference between underfitting and overfitting.
4. How does dropout prevent overfitting?
5. Compare ReLU and Sigmoid activation functions.